Few-Shot Prompt-Based Longitudinal Mood-State Analysis of Bipolar Disorder on Social Media

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Abstract. Bipolar disorder (BD) is frequently misdiagnosed as major depressive disorder, and monitoring mood-state *transitions* over time is critical for timely intervention. Existing social media datasets for BD research typically provide only binary diagnostic labels or per-post mood scores; few capture mood trajectories. Because expert annotation is costly and fine-tuning requires scarce labeled data, we propose a few-shot prompt-based LLM method that requires no task-specific training. The annotation schema is grounded in DSM-5 episode criteria and accompanied by eight synthetic few-shot examples, making the method directly applicable to other BD corpora. The method labels posts at two granularities: per-post mood state and 14-day period-level trends (dominant state, trend direction, change points), enabling trajectory modeling that per-post labels alone cannot support. Post-level classification is externally validated against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with high depressive recall (87.9%) and lower manic-pole recall (35.7%/6.7%). Six error patterns are characterized, including a structural label-text consistency issue at the manic pole that bounds achievable manic recall. Using Gemini 3.1 Pro, we apply the method to 105 self-identified BD users on BD-focused subreddits (1,794 14-day periods, 15,423 posts and comments, April 2019–May 2026), observing depressive-pole predominance broadly consistent with clinical expectations for BD-related online discussion.

Keywords: Bipolar Disorder · Large Language Models · Few-Shot Prompting · Social Media · Clinical NLP · Mood Trajectory

1 Introduction

Bipolar disorder (BD) is characterized by recurrent episodes of mania, hypomania, and depression, affecting 1–2% of the global population [2]. An estimated 17–50% of BD cases are initially misdiagnosed as major depressive disorder (MDD), because patients typically seek help during depressive episodes and may not recognize manic or hypomanic states as pathological [3, 4]. This misdiagnosis leads to inappropriate treatment (e.g., antidepressant monotherapy, which may trigger manic switching) and delays proper intervention.

Social media platforms such as Reddit host BD communities (e.g., r/bipolar, r/BipolarReddit) where users openly discuss symptoms, treatment, and daily functioning. Prior work has used these data for BD and MDD classification [5–7], yet existing datasets provide only binary diagnosis labels (BD vs. MDD) or user-level risk scores. Few resources offer post-level mood state labels tracked over time, limiting computational research on mood trajectories and, in turn, on BD progression and early intervention.

Constructing expert-annotated BD corpora is costly: mood-state labeling requires psychiatric expertise, and the scarcity of labeled data makes fine-tuning approaches difficult to scale. At the same time, recent advances in large language models (LLMs) suggest that they can follow human-written guidelines with reasonable accuracy on text annotation tasks [8]. This motivates our approach: if LLMs can internalize clinical guidelines presented in a prompt, then a carefully designed DSM-5-grounded prompt with synthetic few-shot examples should enable mood-state annotation *without task-specific fine-tuning or labeled training data*, making the method directly applicable to new BD corpora.

Recent work has applied LLMs to mental health NLP tasks [9, 10]; however, whether they can classify BD mood states at expert-level accuracy, especially manic-pole states that are difficult to detect from text, has not been established.

This paper proposes a few-shot prompt-based LLM method for longitudinal mood-state analysis of BD on social media and demonstrates its application to a Reddit cohort. We collect Reddit posts from BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2), targeting users who self-identify as having a BD diagnosis, and annotate them at two granularities: (1) per-post mood state (*Depressive*, *Stable*, *Hypomanic*, *Manic*) and (2) 14-day period-level mood trends (dominant state, trend direction, change points). The annotation schema follows DSM-5 episode criteria and is implemented as an LLM pipeline (Gemini 3.1 Pro). We validate the post-level annotations against the BD-Risk dataset [1] on a held-out, author-disjoint, stratified subset of 145 posts, achieving macro F1 of 0.519 with 87.9% depressive recall and 35.7%/6.7% recall on hypomania/mania. The recall asymmetry across poles reflects both a known model property (manic-side states often manifest through described behaviors rather than affective tone) and a structural label-text consistency issue with the source dataset's manic-pole labels.

Our contributions:

1. **Method:** A DSM-5-grounded few-shot prompt schema for LLM-based mood-state annotation at two temporal granularities (per-post state and 14-day period-level trend), requiring no fine-tuning or labeled training data.
2. **Evaluation:** External validation of post-level state classification against the BD-Risk expert-labeled dataset [1] on a held-out, author-disjoint, stratified subset, with macro F1 of 0.519 (87.9% depressive recall, 35.7% hypomanic recall, 6.7% manic recall), complemented by a zero-shot baseline comparison, a supervised fine-tuning baseline, and a cross-model portability evaluation across five additional LLMs from five providers (macro F1 0.432–0.564).

3. **Demonstration:** Application of the method to 105 self-identified BD users from BD-focused subreddits (1,794 14-day periods, 15,423 posts and comments spanning April 2019 through May 2026), producing longitudinal mood trajectory annotations with distributions aligning with clinical BD literature.

2 Related Work

2.1 Longitudinal Mood Monitoring in BD

Clinical tracking of BD mood trajectories relies heavily on ecological momentary assessment (EMA) and digital phenotyping via smartphone sensors and self-report apps [11, 12], which yield dense mood signals yet require active enrollment and consent, limiting cohort size and external use. Social media offers a complementary unobtrusive source of passive, naturally produced language over months to years for users already discussing their condition; however, public corpora with period-level mood-trajectory annotations remain limited, constraining longitudinal BD method development.

2.2 Social Media in Mental Health Detection

De Choudhury, M. et al. [13] showed that social media signals can predict depression onset. The SMHD dataset [5] covers nine mental health conditions via self-reported diagnoses; Coppersmith, G. et al. [6] established shared tasks for depression and PTSD detection from X (formerly Twitter).

For BD, Sekulić, I. et al. [7] proposed Reddit-based classification and Jagfeld, G. et al. [14] compiled a large BD Reddit corpus; however, both rely on self-reported diagnoses without expert validation [15, 16]. The BD-Risk dataset [1] provides per-post mood labels validated by a psychiatrist and a clinical psychologist; we use it as the gold standard for our post-level validation.

2.3 LLMs for Clinical NLP and Mental Health

Xu, X. et al. [9] evaluated LLMs on mental health prediction from online text; Yang, K. et al. [10] fine-tuned MentalLLaMA for interpretable mental health analysis. Lee, D. et al. [1] found that ChatGPT achieved only an F1 of 0.130 on BD risk detection (vs. 0.578 for their multi-task model), showing that off-the-shelf LLMs struggle with BD-specific tasks. On the more general question of LLM annotation quality, Gilardi, F. et al. [8] show that ChatGPT can match or exceed crowd workers on text annotation tasks; our work follows this line by treating the LLM as an annotator (not a classifier) and grounding its outputs in an explicit DSM-5-derived schema.

Unlike these diagnosis-prediction evaluations, we propose a prompt-based method that *annotates* per-post mood states and period-level trends, and validate the annotations against expert labels; the error analysis (Sect. 4.7.3) characterizes the failure modes the schema must address.

3 Methodology

3.1 Method Overview

The proposed method takes as input a user's posting history from mental-health-related social media communities and produces mood-state annotations at two temporal granularities (per-post state and 14-day period-level trend). Fig. 1 shows the end-to-end flow: candidate identification through LLM-based patient verification, followed by structured annotation with two DSM-5-grounded prompts.

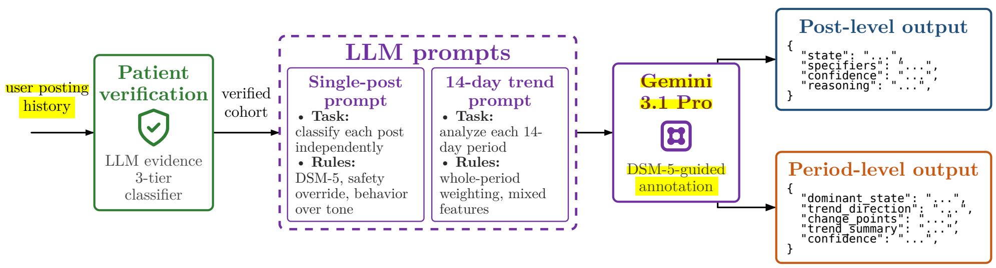

Fig. 1. Annotation pipeline producing structured mood-state labels at two temporal granularities (post-level and 14-day period-level).

Posting to a mental-health-related subreddit is a necessary yet insufficient signal for BD diagnosis: many such posts come from clinicians, family members, or general community participants. To screen the candidate pool, we apply an LLM three-tier classifier (Gemini 3.1 Pro, separate prompt) that scans each author's full posting history and returns `verified` (explicit first-person diagnostic statements, e.g., "I was diagnosed with bipolar II in 2019", or specific treatment/hospitalization narratives), `probable` (consistent self-identification through symptoms, medication, or community-membership tone without an explicit diagnosis statement), or `unverified` (no diagnostic signal). Only the `verified` and `probable` tiers are admitted to the annotation cohort; this matches the inclusion model used in prior BD social-media datasets [7, 14] with stricter per-user evidence gating than a one-post membership rule.

3.2 Annotation Schema

Our annotation schema draws on DSM-5 episode definitions [17], operationalizing them as a structured prompting framework for LLM-based annotation. The rules below were developed iteratively against an error analysis on external expert labels (see Sect. 4.7.3); each clinical-guidance rule named below is the schema's response to a recurring failure mode characterized there. Because BD-Risk posts contain sensitive mental health disclosures, the error analysis illustrates each pattern with a synthetic case rather than verbatim post text, preserving the clinically relevant structure of the original disagreement.

Post-Level State Classification For each individual post (submission or comment), the LLM assigns a categorical mood state from five options:

- **Manic:** Grandiosity, pressured writing (run-on sentences, excessive capitalization), flight of ideas (tangential topic shifts), extreme irritability or euphoria.
- **Hypomanic:** Elevated energy and pace with maintained coherence, social disinhibition, uncharacteristic intensity without psychotic features.
- **Depressive:** Linguistic constriction, absolutist language ("never," "nothing"), high self-focus (first-person pronouns), cognitive distortions, suicidal ideation.
- **Stable:** Balanced emotional tone, metacognitive reflection, proportionate responses, community-supportive language.
- **Uncertain:** Reserved for truly uninterpretable posts; the LLM must attempt classification before resorting to this label.

The framework also supports a `with_mixed_features` specifier (following DSM-5 mixed-features criteria). Before applying this specifier, the prompt requires the LLM to extract an explicit list of opposite-pole symptoms, and only assigns `with_mixed_features` when three or more clear opposite-pole symptoms are documented. This evidence-extraction step prevents the mixed-features specifier from being used as a vague neutral label.

Period-Level Trend Analysis For longitudinal mood trajectory modeling, we partition each user's posting history into consecutive fixed-length periods (default: 14 days). The 14-day window length was set in consultation with clinical advisors and is grounded in DSM-5 episode-duration criteria, which define a major depressive episode as lasting at least two weeks and a manic episode as lasting at least one week [17]; a 14-day window therefore captures the minimum duration of a full depressive episode and allows observation of manic-episode onset and progression. Periods are anchored at the user's first post (day 0) and advance in strict half-open intervals $[t_k, t_k + 14)$; submissions and comments are jointly assigned to the period containing their timestamp. All periods from the user's first to last post are defined; periods without posts receive a `NO_DATA` label rather than being skipped, preserving a continuous time grid for trajectory modeling. Fig. 2 illustrates the segmentation.

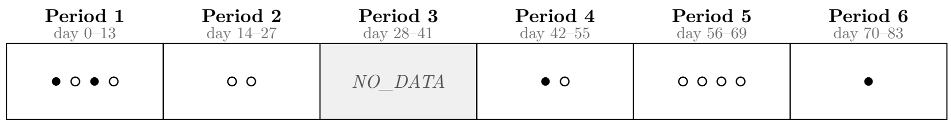

Fig. 2. Period segmentation (illustrative). Submissions (filled circles) and comments (open circles) fall in the period containing their timestamp; empty periods carry the `NO_DATA` label.

For each period containing at least one post, the LLM analyzes the collected posts and produces:

- **Dominant state:** The primary mood state across the period (same five-class set as post-level), aggregated across the posts in the window.
- **Trend direction:** `NO_TREND` (state maintained), `TOWARDS_MANIA` / `TOWARDS_DEPRESSION` (progressive worsening toward the respective pole), or `FLUCTUATING` (alternation without a clear direction).
- **Change points:** Specific dates or events where a mood shift occurred, with pre- and post-shift states documented.
- **Trend summary:** A concise narrative describing the period's trajectory and the evidence supporting the dominant state.
- **DSM-5 specifiers:** `with_mixed_features` when opposite-pole symptoms co-occur within the period (distinguished from sequential fluctuation).

The two granularities support both event-level analysis (e.g., what preceded a state change) and longitudinal trajectory modeling, with the explicit change-point fields enabling change-point detection [18] at the textual level.

Few-Shot Example Construction The post-level prompt is paired with eight synthetic few-shot examples (labeled A–H), written by the authors. Each exercises a schema rule targeting a failure mode observed during development (full analysis in Sect. 4.7.3): A demonstrates *Behavior Over Tone* on retrospective manic-side narration; B and E demonstrate the *SAFETY OVERRIDE* rule with its grandiose-mania exception; C and D contrast *Improvement-Narrative* against *Whole-Post Evidence Weighting*; F counters the default-to-*Uncertain* tendency on short posts; G demonstrates the *Recurrent-Pattern Exception* for substance-triggered hypomania; H exercises *Severity Descriptors* for *Hypomanic*-vs-*Stable* boundaries. Each example provides the input text with the full expected JSON output (including the `opposite_pole_symptoms` evidence list and reasoning), so the model observes both the target label and the evidence chain. The full prompt texts are available in the supplementary repository (see the Appendix).

4 Validation Experiments

4.1 External Validation Against BD-Risk

The BD-Risk dataset [1] comprises 7,346 Reddit posts from 1,025 users, each carrying a psychiatrist-guided mood level label on a 7-point scale (-3 to $+3$). Because the dataset selects users based on an initial MDD presentation (MDD-only and MDD→BD groups), it is structurally enriched for depressive-pole content (89.0% of posts < 0).

The BD-Risk dataset provides only ordinal mood labels; categorical states are not directly annotated. To obtain gold states for evaluation, we derive them from BD-Risk mood labels using the mapping shown in Table 1.

Table 1. Mapping from BD-Risk 7-point mood labels to derived gold states.

BD-Risk mood label	Derived gold state	Notes
$-3, -2, -1$	Depressive	
$0, +1$	Stable	$+1$ = high motivation / positive mood within normal range
$+2$	Hypomanic	Clear manic-side activation without psychosis
$+3$	Manic	Severe manic expression with psychotic features

The full BD-Risk dataset exhibits a heavily skewed mood distribution (89.0% of posts < 0). Because the deployment task is BD risk detection rather than general mood classification, we deliberately oversample manic-pole posts so that per-class metrics on the underrepresented classes are computed with sufficient support.

We separate the labeled posts into two disjoint subsets. A *development* subset (314 posts) is used during prompt design and failure-mode analysis. A *held-out* subset (145 posts) is used exclusively for the evaluation reported below; it is *author-disjoint* from the development subset, drawn by stratified sampling from BD-Risk authors not present in the development subset, with quotas ensuring sufficient per-class support across the four derived gold states (60 *Depressive*, 40 *Stable*, 30 *Hypomanic*, 15 *Manic*). All metrics in Sect. 4.5 are computed on the held-out subset; the development subset is never used to produce reported numbers. Manic-pole gold posts in BD-Risk come almost exclusively from the MDD→BD group, so the held-out manic-side samples are structurally MDD→BD-derived (a limitation discussed in Sect. 6).

4.2 Evaluation Metrics

We report per-class precision, recall, and F1, along with overall accuracy and macro F1. Two accuracy variants are reported: accuracy *excluding Uncertain* treats *Uncertain* outputs as abstentions and removes them from both numerator and denominator; accuracy *including Uncertain* counts *Uncertain* as incorrect, providing a conservative lower bound. Per-class metrics are computed on the excluding-Uncertain basis.

4.3 Evaluation Design

Beyond the main BD-Risk holdout validation, we conduct three additional evaluations to characterize the schema's properties and situate its performance relative to supervised alternatives:

- **Zero-shot baseline comparison:** To quantify how much the structured annotation schema (DSM-5 rules, few-shot examples) contributes beyond the LLM's base capability, we re-evaluate the same model with a minimal zero-shot prompt containing only the task definition and output format.
- **Cross-model portability evaluation:** To test whether the schema generalizes beyond the primary annotator, we evaluate the full schema with five additional LLMs spanning five providers (Gemini 3.5 Flash, Claude Opus 4.8, DeepSeek V4 Pro, GLM-5.1, and GPT-5.5), characterizing schema portability, per-model manic-pole behavior, and the impact of provider-level content-policy differences on annotation feasibility.
- **Supervised fine-tuning baseline:** To assess whether task-specific fine-tuning with labeled data outperforms the proposed few-shot approach, we fine-tune ModernBERT-base [19] on progressively larger subsets of the 314-example training pool ($n \in \{50, 100, 200, 314\}$) and evaluate on the same 145-example held-out set, providing a direct comparison under identical test conditions.

4.4 LLM Configuration

We use Gemini 3.1 Pro [20] through the official API with structured JSON output; the model was selected for its large context window and native structured-output generation. Each post is processed independently with the full annotation schema and the few-shot examples described above as the system instruction; the model returns a JSON object with `state`, `opposite_pole_symptoms`, `specifiers`, `confidence` (High/Medium/Low), and `reasoning` fields, where `opposite_pole_symptoms` carries the explicit evidence list required before `with_mixed_features` can be assigned (see Sect. 3.2). For period-level annotation the LLM additionally returns `trend_direction`, `change_points`, and a `trend_summary` narrative, with `confidence` on a 0–1 scale. We use the Gemini-recommended default temperature of 1.0; Google's documentation for Gemini 3 models advises against lower values, as they may cause looping or degraded performance on complex reasoning tasks. The model is not fine-tuned.

4.5 Validation Results

We first validate the post-level state classification against BD-Risk expert labels (Table 2: per-class metrics with macro-aggregated summary); because the held-out subset is intentionally stratified across classes, macro F1 is the primary metric.

Table 2. Per-class metrics with macro summary on the held-out subset (n=145; 138 excluding 7 *Uncertain*). Accuracy excluding/including *Uncertain* = 65.9 % / 62.8 %.

State	Precision	Recall	F1	Support
Depressive	0.630	**0.879**	**0.734**	58
Stable	0.659	0.784	0.716	37
Hypomanic	**0.833**	0.357	0.500	28
Manic	**1.000**	0.067	0.125	15
Macro avg	*0.781*	*0.522*	*0.519*	*138*

Depressive recall is high (87.9%) and *Stable* recall is moderate (78.4%), while *Hypomanic* and *Manic* recall remain low (35.7% and 6.7%), indicating that the LLM correctly recognizes most depressive and stable posts while missing a majority of manic-pole cases. The dominant error flow runs from the manic pole to *Depressive*: among the 30 gold-*Hypomanic* posts, 12 are predicted *Depressive* and 6 *Stable*; among the 15 gold-*Manic* posts, 11 are predicted *Depressive* and 2 *Stable*. These errors are concentrated on manic-pole posts whose activation is masked by negative tone, a pattern that the current prompt does not fully resolve. The Manic-to-Depressive error flow is a recurring pattern with a likely label-text origin that we discuss in Sect. 6.

4.6 Supervised Fine-Tuning Baseline

To contextualize the few-shot LLM results, we compare against a supervised baseline. We fine-tune ModernBERT-base [19] (149M parameters) on the same BD-Risk training pool using progressively larger labeled subsets ($n \in \{50, 100, 200, 314\}$), evaluating on the identical 145-example held-out set. Each tier is trained for 10 epochs with a learning rate of 2×10^{-5} and a maximum sequence length of 2,048 tokens; for tiers 50–200, we report the mean and standard deviation over five random training-set samples (seeds 42–46), while tier 314 uses all available training examples and reports variance over five random initializations. Table 3 summarizes the results alongside the LLM conditions.

Table 3. Supervised fine-tuning baseline vs. LLM annotation on the held-out subset ($n = 145$). ModernBERT results report mean ± std over 5 seeds. The few-shot LLM uses eight synthetic examples and requires no labeled training data.

Method	Training data	Macro F1	Δ vs. few-shot
ModernBERT	$n = 50$	0.306 ± 0.022	−0.213
ModernBERT	$n = 100$	0.337 ± 0.018	−0.182
ModernBERT	$n = 200$	0.372 ± 0.019	−0.147
ModernBERT	$n = 314$	0.398 ± 0.018	−0.121
Gemini 3.1 Pro	zero-shot	0.459	−0.060
Gemini 3.1 Pro	8 few-shot	**0.519**	—

ModernBERT macro F1 increases monotonically with training size ($0.306 \rightarrow 0.398$), yet even at the maximum available training size ($n = 314$), it falls short of Gemini few-shot (0.519) by 0.121 and barely approaches the Gemini zero-shot level (0.459). We additionally explored training for 15 and 20 epochs on the full 314-example set: 15 epochs reaches 0.454 ± 0.038 and 20 epochs 0.445 ± 0.010, narrowing the gap yet remaining below the few-shot result in both cases.

Per-class analysis reveals that ModernBERT shares the same manic-pole difficulty observed in the LLM results (Sect. 4.5): across the five tier-314 runs, *Manic* recall averages 0.04 (2 of 5 runs produce zero *Manic* recall), while *Depressive* and *Stable* F1 average 0.51 and 0.62 respectively. This parallel suggests that the manic-pole limitation is rooted in the BD-Risk label–text relationship (Sect. 4.7.3, Pattern 1) rather than in the choice of model architecture.

Labeling 314 training examples for BD mood-state classification required psychiatrist-guided annotation; the few-shot LLM approach outperforms this baseline using only eight synthetic prompt examples and no manual labeling.

4.7 Interpretation of Validation Results

The LLM assigned *Uncertain* to 7 posts (4.8%), abstaining when post content was insufficient for state assessment, consistent with the prompt's explicit instruction to prefer abstention over forced classification. *Uncertain* emissions are distributed across gold classes (2 *Depressive*, 3 *Stable*, 2 *Hypomanic*, 0 *Manic*), with no strong concentration on a single pole.

Schema Contribution: Comparison with a Zero-Shot Baseline Following the design in Sect. 4.3, we compare the full schema against a minimal zero-shot prompt on the held-out subset. The zero-shot prompt retains only the task definition (post → one of five states) and the output JSON fields; all DSM-5 rules, *SAFETY OVERRIDE*, *Severity Descriptors*, and few-shot examples are removed. Table 4 reports the side-by-side metrics.

Table 4. Schema contribution on the held-out subset (n=145). Both runs use Gemini 3.1 Pro with identical JSON output; the only variable is the system prompt (minimal zero-shot versus the full schema).

Metric	Zero-shot	Full schema	Δ
Accuracy (incl. Uncertain)	51.7%	**62.8%**	+11.1 pp
Accuracy (excl. Uncertain)	63.6%	65.9%	+2.3 pp
Macro F1	0.459	**0.519**	+0.060
Uncertain count	27	**7**	−20
DEPRESSIVE F1	**0.738**	0.734	−0.004
STABLE F1	0.600	**0.716**	+0.116
HYPOMANIC F1	0.500	0.500	± 0.000
MANIC F1	0.000	**0.125**	+0.125

The schema's primary effect is reducing abstention and anchoring border-class decisions: *Uncertain* emissions drop 4× (27 → 7) and *Stable* F1 improves by +0.116 (driven by the Severity Descriptors' explicit rule that mild positive activation falls within *Stable*), which explains why the accuracy gain including *Uncertain* (+11.1 pp) far exceeds the gain excluding it (+2.3 pp). *Depressive* and *Hypomanic* F1 are unchanged, and *Manic* remains poorly recalled in both runs (0/15 zero-shot vs. 1/15 schema, counting *Uncertain* as incorrect), indicating that the manic-pole limitation is structural (label-text consistency, Sect. 6) rather than a limitation resolved by this richer prompt.

Cross-Model Schema Portability To test generalization beyond Gemini 3.1 Pro, we apply the identical prompt and JSON format to five additional LLMs from five providers on the same 145-post subset. Table 5 reports the four with complete annotations; GPT-5.5, which declined a majority of posts, is reported in-text.

Table 5. Cross-model evaluation on the held-out subset ($n = 145$), identical 8-example few-shot prompt. Macro F1 is over the four target classes, excluding *Uncertain* (see Sect. 4.5).

Model	N	Macro F1	Dep Rec	Hyp Rec	Man Rec	Unc
Gemini 3.5 Flash	145	**0.564**	0.850	0.379	0.200	3
Gemini 3.1 Pro	145	0.519	0.879	0.357	0.067	7
Claude Opus 4.8	145	0.535	0.932	0.429	0.067	7
GLM-5.1	145	0.456	0.881	0.286	0.000	4
DeepSeek V4 Pro	145	0.432	0.833	0.241	0.000	2

The comparison yields three observations. First, the schema produces valid structured output across all five tabulated models with zero refusal (macro F1 0.432–0.564), indicating it is not provider-specific; GPT-5.5, by contrast, refused 103 of 145 posts under the identical prompt and reached 0.710 only on the 42 it answered, so we report it separately as not directly comparable. Second, manic-pole underdetection is consistent across families (zero *Manic* recall for DeepSeek V4 Pro and GLM-5.1; 6.7% for Claude Opus 4.8 and Gemini 3.1 Pro; 20.0% for Gemini 3.5 Flash), reinforcing the structural interpretation (Sect. 6) that the difficulty stems from the BD-Risk label–text relationship rather than any single model. Third, *Hypomanic* recall varies more (0.241–0.429), consistent with manic-side detection hinging on the model's ability to read behavioral cues over affective tone (Pattern 1, Sect. 4.7.3).

Gemini 3.5 Flash achieves the highest macro F1 (0.564) among models evaluated on the full held-out subset, exceeding Gemini 3.1 Pro by 0.045. Because the corpus was annotated with Gemini 3.1 Pro prior to this cross-model comparison, the existing annotations remain unchanged; the result indicates that schema performance improves with model capability and that re-annotation with newer models could improve manic-pole coverage.

GPT-5.5 declined to classify 103 of 145 held-out posts (71.0%) with a verbatim refusal (`"I'm sorry, but I cannot assist with that request."`), concentrated on posts containing explicit self-harm or suicidal content. The prompt's clinical-research framing did not overcome the refusal. On the 42 posts GPT-5.5 did classify (skewed away from depressive-crisis content: 16 *Depressive*, 10 *Stable*, 12 *Hypomanic*, 4 *Manic*), it achieved macro F1 of 0.710, driven primarily by higher *Manic* recall (2/4). The subset selection is the dominant effect: GPT-5.5 systematically filtered out the hard depressive-crisis posts that drive most of Gemini's error rate, so the macro-F1 comparison overstates GPT-5.5′s effective competence. A 71% refusal rate renders GPT-5.5 infeasible as a stand-alone annotator for psychiatric corpora regardless of intrinsic capability.

Error Analysis We characterize six failure modes identified through manual analysis of LLM-vs-BD-Risk disagreements on the development subset. For each pattern, we name the schema rule designed to mitigate it and note whether the rule resolves the pattern on the held-out subset or whether it remains a residual error. Manic-pole posts misclassified as depressive or stable remain the dominant residual failure mode (quantified in Sect. 4.5).

Pattern 1, the dominant manic-side error, is the LLM ignoring retrospective behavioral cues. When users describe manic-episode behaviors (impulsive spending, aggressive confrontations, hyperactivity) in a retrospective post written with remorse or self-blame, the LLM anchors on the *current emotional tone* rather than the *clinical significance of the described behaviors*, and predicts *Depressive*. The schema's *Behavior Over Tone* rule directly targets this conflation; however, it remains the dominant residual error on the held-out subset, indicating that the rule reduces the pattern without fully eliminating it. *Synthetic case:* a user recounts a week of reckless spending and impulsive decisions in a tone of deep regret and self-condemnation; gold is *Hypomanic* (the behaviors are hallmark manic symptoms), the LLM predicts *Depressive* (anchored on the self-deprecating tone).

Pattern 2 is the conflation of mixed features with neutrality. When a post carries symptoms from both poles simultaneously (e.g., severe sleep disruption and inability to concentrate alongside aggressive outbursts), the LLM sometimes treats the coexistence as cancellation and defaults to *Stable*. Under DSM-5 [17], mixed features should instead surface as a `with_mixed_features` specifier on the dominant pole. The schema's mixed-features rules reduce this conflation, though residual instances remain when symptom signals are subtle. *Synthetic case:* a user writes that they have not slept in three days, cannot focus at work, and snapped at their partner, all in the same post; gold is *Hypomanic* with mixed features (decreased need for sleep alongside dysphoric irritability), the LLM predicts *Stable* on the reasoning that the signals "balance out".

Pattern 3, a safety-critical case, is calm writing style masking suicidal ideation. Some posts express suicidal ideation ("I want to die") in calm, reflective, or educational prose; the LLM reads the linguistic register as stable and classifies accordingly, while the gold state is *Depressive* (−3). Calm writing does not rule out suicidal crisis, and the schema's explicit *SAFETY OVERRIDE* rule forces *Depressive* whenever crisis-level language is present regardless of surrounding tone. On the held-out subset, no *Stable* or *Uncertain* predictions were observed for posts with explicit suicidal content, suggesting that the rule addresses this pattern on the evaluated subset. *Synthetic case:* a user posts a coherent essay about public misconceptions of depression; embedded in the second paragraph, in the same calm register, is a single sentence noting that the writer quietly thinks about not waking up most mornings.

Pattern 4 is end-of-post hope overriding pervasive impairment. Posts describing severe functional impairment (academic collapse, inability to maintain daily routines, social withdrawal) sometimes close with a single hopeful sentence (e.g., "my therapist said maybe I shouldn't give up"), and the LLM anchors on this terminal positive signal (consistent with a recency bias) to predict *Stable* while the gold label tracks the pervasive impairment throughout. The schema's *Whole-Post Evidence Weighting* rule addresses this by instructing the LLM to weigh the dominant clinical picture over terminal sentiment; the zero-shot comparison (Table 4) shows that the full schema improves *Stable* F1 by +0.116, consistent with reduced over-assignment of *Stable* to impaired posts. *Synthetic case:* a user reports missing every class for a month, eating almost nothing this week, and losing the ability to reply to friends, closing with "maybe tomorrow will be different"; gold is *Depressive* on the pervasive impairment, the LLM predicts *Stable*.

Pattern 5 concerns substance-induced activation versus endogenous mood. When users describe mood elevation explicitly attributed to substances (e.g., caffeine-induced euphoria described as feeling "on top of the world"), the LLM may interpret these cues as hypomanic mood, while gold labels distinguish acute pharmacological activation from endogenous baseline. The schema's *Substance vs. Endogenous Mood* rule addresses this boundary and also covers the *Recurrent-Pattern Exception* where an unusual reactivity to a common substance is itself bipolar-spectrum. This pattern is rare in the held-out subset; the rule's primary effect is preventing false *Hypomanic* predictions on substance-attributed posts. *Synthetic case:* a user writes that after a fourth coffee they suddenly feel "unbeatable and ready to overhaul" their apartment, attributing the surge to caffeine and expecting an evening crash; gold is *Stable* (acute, externally caused, self-labeled), the LLM predicts *Hypomanic* on the surface cues.

Pattern 6 is *Uncertain* masking embedded clinical signals. *Uncertain* functions appropriately in the large majority of cases; the failure described here is rare. The LLM correctly identifies the post as metacommentary or informational, then fails to surface clinically significant content embedded inside. *Synthetic case:* a user writes a meta-commentary criticizing how recovery is portrayed online and, mid-argument, notes parenthetically that they have been thinking about ending things;

gold is *Depressive*, the LLM defaults to *Uncertain* on the meta-discussion register. The *SAFETY OVERRIDE* rule mitigates this pattern by forcing *Depressive* on crisis-level language regardless of overall framing. Like Pattern 3, the rule appears effective on the held-out subset: no crisis-level posts were classified as *Uncertain*.

5 Longitudinal Demonstration: Period-Level Mood Trends

We continuously crawl three BD-focused subreddits (r/bipolar, r/BipolarReddit, r/bipolar2) via the Reddit API, retrieving each active author's full posting history (submissions and comments) with periodic re-crawls to capture ongoing activity. After the patient verification described in Sect. 3.1, the verified+probable cohort comprises 115 of 124 candidate authors. Users with fewer than two 14-day periods containing posts (i.e., insufficient longitudinal span for trend analysis) are excluded. The remaining 105 users contribute 2,611 submissions and 12,812 comments spanning April 2019 through May 2026, yielding 1,794 valid analysis periods.

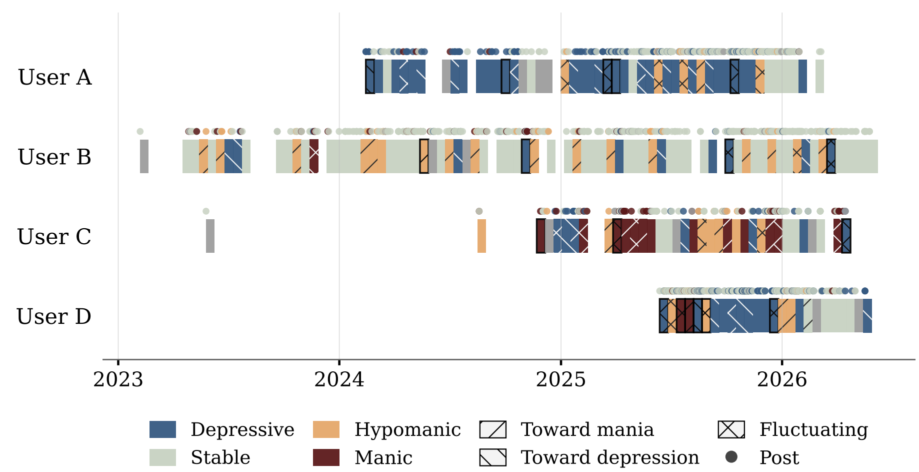

Fig. 3. Mood trajectories for four anonymized users (A: depressive with fluctuation, 49 periods; B: hypomanic-leaning with frequent manic transitions, 74 periods; C: manic-dominant with rapid cycling, 36 periods; D: dense posting with high mixed-features incidence, 25 periods). Each bar is a 14-day period; a thin black border marks `with_mixed_features`. Dots above bars are post-level annotations colored by post state. State and trend encodings follow the legend.

Most 14-day windows show NO_TREND (83.9%); TOWARDS_DEPRESSION and TOWARDS_MANIA account for 5.9% and 4.3% respectively. *Stable* and *Depressive* states dominate (42.6% and 31.4% of periods), while manic-pole states are less frequent (*Hypomanic* 8.8%, *Manic* 3.0%). These distributions are interpreted in Sect. 6.

Table 6. Period-level distributions on the 105-user cohort ($n = 1,794$): trend directions (left) and dominant states (right). The with_mixed_features specifier was applied to 84 periods (4.7%).

Trend direction	Periods	%
NO_TREND	1,506	83.9%
FLUCTUATING	106	5.9%
TOWARDS_DEPRESSION	105	5.9%
TOWARDS_MANIA	77	4.3%

Dominant state	Periods	%
Stable	765	42.6%
Depressive	564	31.4%
Hypomanic	157	8.8%
Manic	54	3.0%
Uncertain	254	14.2%

Submissions ($n = 2,611$) carry the majority of polar-state labels: 36.8% *Depressive*, 10.6% *Hypomanic*, 3.8% *Manic*, 46.6% *Stable*, and 2.2% *Uncertain*. Comments ($n = 12,812$) are predominantly conversational replies labeled *Stable* (88.1%), with polar states comprising only 8.9% of comments. The submission–comment asymmetry and its implications are discussed in Sect. 6.

6 Discussion

From the trend distribution (Sect. 5), four observations stand out. (1) TOWARDS_MANIA and TOWARDS_DEPRESSION trends are rare yet among the most clinically significant signals, as they mark episode onset where intervention has the most impact. (2) FLUCTUATING periods may correspond to rapid cycling or mixed presentations that single-post labels cannot capture. (3) Post-level states and period-level trends together enable hierarchical modeling: predicting the next period's trajectory from the sequence of post-level features. (4) The dominant-state distribution preserves sufficient manic-pole representation (*Hypomanic* 8.8%, *Manic* 3.0%), in contrast to MDD-selected cohorts (BD-Risk: 89.0% depressive-pole); this balance matters for downstream models that must distinguish manic-pole from depressive states. The trend distributions broadly match long-term BD cohort studies [2] while reflecting the selection and posting biases of Reddit communities; direct external validation would require period-level expert annotations. A complementary direction is to model the absence signal in NO_DATA periods: reduced posting is sometimes associated with depression in digital-phenotyping research [11], yet the relationship is heterogeneous, so absence modeling would require posting-frequency baselines beyond a text-based pipeline.

Submissions and comments differ sharply: 51.2% of submissions carry a polar state vs. 8.9% of comments, which are dominated by *Stable* (88.1%). Submissions are longer-form disclosures while comments are short replies. Two downstream implications: a per-post classifier on a comment-heavy corpus is likely to appear over-confident on *Stable*, so per-content-type metrics are preferable to a single aggregate; and trajectory models should either up-weight submissions or rely on the period-level dominant-state annotation (which already aggregates across content types within the window) as the primary trajectory signal.

The LLM achieves 87.9% recall for *Depressive* yet only 35.7% for *Hypomanic* and 6.7% for *Manic* on the held-out subset. Two factors compound this asymmetry. First, a model property: depressive language has stereotypical surface markers (negativity, self-focus, hopelessness), while manic-pole states often manifest through *described behaviors* (spending sprees, reduced sleep need, grandiose plans) narrated in any tone, and the LLM reads tone rather than the clinical significance of the behaviors described (see Pattern 1 in Sect. 4.7.3). Second, a label–text consistency issue in the BD-Risk annotation rule: as specified by Lee, D. et al. [1] (Section 3.2), "posts exhibiting both manic and depressive moods are regarded as manic moods", an asymmetric tie-breaker that elevates any mixed manic+depressive post to the manic side. Combined with the dataset's MDD→BD selection criteria, this yields gold-label *Manic* (+3) posts whose textual content matches BD-Risk's own definition of −3 ("extreme anxiety and having suicidal thoughts"). A single-post LLM applying clinical-safety priors (classifying explicit self-harm content as *Depressive*) will systematically underperform on these posts, since the only signal it has is the very content the labeling rule overrode. The cross-model evaluation (Sect. 4.7.2) provides direct evidence for the structural interpretation: across six LLMs from five providers, *Manic* recall ranges from 0.0% (DeepSeek V4 Pro, GLM-5.1) to 20.0% (Gemini 3.5 Flash), and the supervised ModernBERT baseline averages 4.0% (Sect. 4.6); all architectures converge on the same manic-pole floor. The Manic-recall ceiling in Table 2 thus reflects a structural mismatch, not solely a model limitation (downstream implications in Limitations).

7 Limitations

The proposed method and corpus are subject to constraints that we group into evaluation scope, gold-state derivation, manic-pole interpretability, cohort framing, and release-time reliability.

The cross-model evaluation (Sect. 4.7.2) confirms schema portability across five additional LLMs (macro F1 0.432–0.564 on the full 145-post holdout), yet the corpus itself was annotated exclusively with Gemini 3.1 Pro; per-class reliability estimates thus remain Gemini-specific. The quantitative validation is also at the post level only, because few per-post expert-labeled BD datasets at sufficient scale are available for research use (BD-Risk was obtained through a formal data request and ethics review, and comparable shared-task corpora face similar access barriers); period-level trends, a key contribution of the method, are not externally validated.

Gold states are derived from BD-Risk ordinal mood labels via a deterministic mapping (Table 1) rather than being directly annotated as categorical states. This introduces boundary imprecision (e.g., mood label +1 may reflect mild hypomania rather than stable mood, and an ordinal intensity score need not match a categorical clinical judgment), and mapping the ordinal labels (inter-expert Krippendorff's $\alpha = 0.87$) to categories amplifies disagreement at boundaries. Some apparent misclassifications may thus be mapping artifacts, and reported accuracies should be read as lower bounds on the schema's true reliability.

Two factors limit how the Manic-pole numbers can be read. First, the held-out *Manic* class is small ($n = 15$, because the entire BD-Risk dataset contains only 28 mood-+3 posts); the 95% confidence interval around the Manic-F1 estimate is wide (± approximately 13 percentage points), and small Manic-F1 differences should not be interpreted as significant. Second, the Manic-recall ceiling of 6.7% is bounded by a structural mismatch between BD-Risk's asymmetric tie-breaking rule and the textual evidence available to a per-post classifier (full analysis in Sect. 6); recovering label-text consistency would require either re-annotation under text-only criteria or a longitudinal evaluation protocol that supplies the temporal context the annotators had access to.

Patient verification is LLM-based, screening for self-disclosed BD diagnosis in a user's posting history rather than providing clinical confirmation; it follows prior BD social-media datasets [7, 14] with stricter per-user evidence gating than a one-post membership rule. The cohort may include `verified` or `probable` users describing an unconfirmed diagnosis and conversely exclude users with BD who never disclose it, and Reddit is a self-selected, asynchronous channel whose relationship to clinically observed mood states requires further study. Downstream uses requiring clinician-confirmed status should treat the cohort as an LLM-screened self-identified sample, not a clinical cohort.

All annotations are produced entirely by the LLM without manual spot-checking. Per-class reliability decreases from *Depressive* through *Hypomanic* to *Manic* (see Table 2); *Manic* labels should be used with caution given both the small support and the manic-pole interpretability constraints above.

8 Conclusion

The validation separates the proposed method's two failure sources: 87.9% depressive recall suggests that the schema captures dominant depressive-pole markers with reasonable coverage, while the 6.7% manic recall appears strongly influenced by a structural label-text consistency issue at the manic pole rather than by prompting alone (see Sect. 6). Because no external longitudinal ground truth exists at the period level, the corpus's 1,794 trajectories are validated only indirectly through their post-level constituents. Three near-term priorities follow: (1) expert annotation of period-level trends to enable direct longitudinal validation; (2) a stratified human-in-the-loop audit across mood states, confidence levels, and trend directions, reported as inter-annotator agreement; and (3) re-annotation of the corpus with a higher-performing model (e.g., Gemini 3.5 Flash, which achieved 0.564 in the cross-model evaluation) to improve manic-pole coverage, combined with targeted prompt revision for the behavioral-cue recognition failure documented in Pattern 1 of Sect. 4.7.3. The resulting corpus is intended for computational mental health research and should not be used as a clinical diagnostic tool.

9 Ethical Considerations

This study involves analysis of publicly posted social media content discussing sensitive mental health experiences. The study protocol was reviewed and approved by the Research Ethics Committee of the University of Tsukuba (approval no. 25-188). We additionally adhere to Reddit's privacy policy and social media research ethics guidelines [15].

Before public release, all post content undergoes LLM-based de-identification across five risk-ranked PII categories (identifiers, quasi-identifiers, contact information, linkage codes, personal identification codes), each detected span replaced by a category-specific placeholder. Unlike rule-based or NER approaches, the LLM also detects accumulated long-span quasi-identifiers (individually innocuous details such as occupation, location, and family structure that jointly narrow identification to one person) while preserving clinically relevant content (medications, diagnoses, symptoms, relative temporal expressions). The full taxonomy and prompt are provided in the supplementary repository.

The dataset retains only text content and temporal information necessary for mood-state annotation; author usernames are replaced with anonymized identifiers, and subreddit membership and post metadata that could facilitate re-identification are excluded from the published dataset. To reduce search-based re-identification risk, released timelines use relative temporal offsets rather than exact posting times. Because LLM-based de-identification may miss residual quasi-identifiers, public release is conditioned on a pre-release privacy audit of a stratified sample. The dataset is intended solely for computational mental-health research and must not be used for re-identification, commercial profiling, or clinical decision-making without appropriate expert oversight.

Appendix

Four system prompts drive the pipeline: (A) the post-level annotation prompt with the full DSM-5-grounded schema and eight synthetic few-shot examples; (B) the 14-day period-level trend prompt; (C) the patient verification prompt; and (D) the de-identification prompt. A minimal zero-shot variant of prompt (A) is used for the schema contribution comparison (Sect. 4.3). Full prompt texts, pipeline source, and evaluation scripts are released at https://anonymous.4open.science/r/bd-state-annotation/.

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